

Colossal Tech: How AI capex is shaping the economy

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Podcast transcript

John Przygocki: Welcome to Talking Markets with Franklin Templeton. I'm your host, John Przygocki from the Franklin Templeton Global Marketing Organization. Today, I'm joined in the studio by Matt Moberg, Senior Vice President and Portfolio Manager with the Franklin Equity Group. Matt oversees several strategies focused on capturing innovation-driven growth. He has been managing money at Franklin Templeton for close to 30 years, and brings a distinctive blend of expertise in history, accounting and finance to his role. Matt, welcome to the show.

Matt Moberg: Thank you, John, and very nice to be here and be able to speak to you in this format. I've had not many opportunities to speak on podcasts, so this should be a lot of fun.

John Przygocki: Fantastic. It's great to see you, and I'm real happy to have another colleague from the Franklin Equity Group on the show. A couple months ago, we had Chief Investment Officer Jonathan Curtis on the Talking Markets platform, and we had a wonderful conversation about AI [Artificial Intelligence]. I'm really excited to connect with you today. So let's jump right in.

Matt, you oversee more than \$30 billion in equity assets with a focus on growth potential of innovation. You studied history as an undergrad. How has your study of history shaped your perspective as an investor focused on identifying innovation-driven transformation?

Matt Moberg: I appreciate that. I received a BA in history in college, which, just to be clear, would only make me a hobbyist or an amateur historian. After college, I received my CPA [Certified Public Accountant] and learned accounting. And I think the combination of these two disciplines has been really helpful in my career.

That said, on history in particular, it's been very useful for my career as a growth investor. I would say there are two main forms of knowledge I rely on when investing: lived experience and historical analogy. On the topic of lived or direct experience, one of the things I've always loved about this industry is the more experience you have with the markets, all things equal, the better investor you should become.

In this industry, experience is measured in bear markets and not in years. So at this point in my career, I have over 25 years of experience. In my opinion, that experience matters. Just like in investing, in which time is such an important variable to understand total absolute returns, I think it's important in this industry to have experienced multiple scenarios. But I also think, more importantly, experience in this industry is measured in bear markets. Those are the most difficult and important times. And I have experienced three bear markets, none of which have been the same. All of which I hope have been helpful to making me a better investor.

Where history for me is so helpful is when we enter into a period that escapes our collective memory. When we use phrases like "unprecedented times" or "this has never happened before," what's really being said in those headlines is this hasn't happened in our lifetime. Actually, most things have happened before, or at least we can draw useful analogies to some other period of history, often outside of our living memory. So, war, starvation and plagues, those have been the three great killers in history and also major events in history. So it's intuitive that when faced with one of these in the present, it's useful to consult the past.

COVID is a great example of this. When COVID broke out, we chose to study other plagues. For example, we looked at the influenza pandemic of 1918 very carefully. And if you look at the photographs from that period, it's remarkable. They look so similar to what we went through in 2020. Face masks and quarantines. In fact, the Spanish flu of 1918 was actually much deadlier. It killed about 50 million people, primarily in Europe. And compare that to the 1.3 million estimated deaths from COVID in the US. Certainly, that period of time in 1918 was much more deadly. But like COVID, you know, the influenza pandemic of 1918 lasted about two years. It came in 3-4 waves, and each wave was more

contagious but also weaker.

What's also informative to us is what happened after the influenza pandemic ended. Western society launched right into the Roaring '20s, which is a period of great prosperity. In fact, this sort of economic upturn following a pandemic is not uncommon. The bubonic plague, that happened even 400 years before the Spanish flu. Afterwards, it made labor so expensive that it led to bargaining power for the surviving laborers that was so high that it ended feudalism in many parts of Europe. And it launched the Renaissance and then the Reformation, one of the most technologically and artistically productive periods in human history.

So when we were in the middle of COVID, it's helpful not to think only about what happens during, but also after, the pandemic. And this helps us get comfortable making that call that this really scary COVID moment might only last for a couple of years, and that the economy could very likely boom coming out of it, and we were able to position the portfolio accordingly.

Another common way that history helps us in investing is helping us simply to put things into perspective. It's pretty common for each new generation to think that they are living in the end of times. Today is certainly no exception. There is a lot of technological change and innovation right now. We are excited to be ending the third and entering into the fourth Industrial Revolution, and our social and political orders are experiencing upheavals along with it. It's fractious, and a lot of people are worried. But again, no one today alive remembers the last Industrial Revolution. And if you study that period, you begin to realize that the innovations that are happening today isn't wildly out of scale or scope. The social changes during that time were arguably far greater than today. When we went from 95% rural to 95% urban living, those lives changed massively, and it was very profound.

And you can go back even further to the founding of the United States. Benjamin Franklin himself was worried about the amount of slander in newspapers, and he thought that the politicians would slander so frequently that he was worried the Republic might not survive. And obviously, that didn't happen.

So, the last point I will make that is important, makes the job so rich, is that we as investors need to learn the right lessons from history. For example, it would be easy to leave the tech bubble of 1999 and decide never to invest in technology. That would be a pity, because that's been the sector that drove investor returns the following quarter century. Similarly, after the housing crisis in 2008, you wouldn't want to take the wrong lesson from that experience and say, "Well, I'm never going to buy a home." Since 2008, homeownership has been a wonderful asset class.

So, understanding the right lessons from history is part of what makes this job so intellectually interesting. And perhaps even more importantly, it helps inform the level of risk I'm willing to take as a portfolio manager in uncertain scenarios.

John Przygocki: Matt, that is wonderful perspective. In your answer, you brought up the terms innovation and technology. So let me go there. These days, people often think of innovation as synonymous with technology or even AI. But I think you have a much broader view. Can you share with our audience how you define innovation?

Matt Moberg: There are two definitions of innovation that I like, and I'll describe them both. First, I think we can think of innovation as the act of bringing something new into the marketplace. Where we focus most of our time in the market is where there was no product and now there is a new product. I really like the Peter Thiel definition, where he would say, you went from 0 to 1. We like to focus on this area because it's the place in the market with the greatest inefficiencies. But notice what's omitted from that definition. It's not tech and it's not AI. You know, vocabulary changes over time, and it can have important differences over that period.

So, the word technology in the 60s and 70s was a word like science. And like the word science, it was fairly universal. Today, technology has a legal definition. It's recognized by the SEC [Securities and Exchange Commission] to have a very specific meaning. For example, if a new fund is founded in the US with technology in the name, under SEC naming rules, it would have to be greater than 80% tech, which is really just hardware, software and semiconductors.

However, for us it has always been a more general definition. One of the things that's great to see today is, although tech has certainly outperformed in the past, that new industries are starting to see massive new companies dominate them. And that's in terms of new companies entering new industries.

So today, over half of the Mag Seven is not even classified as technology. Google and Meta are ad companies, and they're classified under media and entertainment, which to us makes sense. They're competing for attention, just like Disney and videogame makers. You know, Tesla, a car company is competing against Ford and GM. Amazon of course, is a retailer competing against Walmart, Costco. And it's really great to see this broadening out.

However, we're starting to see it move beyond just those large companies. Most fintech companies are showing new innovations in the buy-now-pay-later, in blockchain and cryptocurrencies and payments. That area's classified all under finance. Within energy, we're seeing great advancements in hydrogen energy production. Nuclear is starting to pop up as some of the larger companies in the industrial sector. If you combine autos and transportation together, you would see that ridesharing, autonomous driving and food delivery are new companies that are entering into the top 10 in terms of market cap for those sectors. And finally, space, drones and AI companies are starting to crack into aerospace and defense sectors after decades of stagnation in terms of new names within that space.

So, the reason why this is so exciting [is] because it shows that the market is really dynamic. And although the top 10 in tech and comp services and retail may have had many new companies, we're starting to see that in other sectors and new sectors of the economy, such as energy, industrials and transportation.

We view this as very healthy and a confirmation in our view that all industries will be affected. And we're seeing that wealth creation of new innovations broaden out to the entire economy. We wrote a paper recently that said "Make room Mag Seven, more mega caps are coming." And at that time, there were seven mega caps, defined as greater than a trillion dollars in market cap. And today there are 10.

The second way we think about innovation is in a simple formula. And we wrote about this in a recent paper called "Production," in which we argue that one of our favorite definitions over the past 20 years for innovation is the formula, innovation equals invention times distribution. And while this has really held true while information moats were being created and the company's assets were basically intangible, today we are seeing that transition and we think the proper definition of innovation will become innovation is invention times production, where actually producing goods is starting to become increasingly important.

John Przygocki: So Matt, you just mentioned there are a couple papers that you have recently published. I want to ask you about a third. You've recently written a piece that really put into context what you refer to as colossal tech and the massive investments in artificial intelligence. Can you explain what inspired you to write the paper, and then also provide some detail on why you think it's important to focus attention on these investments?

Matt Moberg: The paper was actually written in response to many client calls we were receiving showing concern over the economy. There was a lot of fretting over tariffs and tax rates and interest rates. In our opinion, one of the hardest things about being a portfolio manager is not only being able to list the issues and understand them but then being able to understand which one of those is the most important.

In our view, economists were weighing all these traditional metrics, but there was one metric that was more important to understanding where the economy was headed at that time and still is. And that is the capex from builders of AI infrastructure. Shorthand, we would say those mega-cap companies. And it may be the most important metric to measure for the future of the health of the economy in the near term. And that's simply because that number is just so large and growing.

And in that paper, what we highlight is, you don't even need to believe in AI applications, which we think, by the way, would be wrong. We think the AI applications will be very good. But what you could instead track is simply what the AI spending is now. And it is massive.

You basically have the largest companies in the world, the largest companies the world has ever created in history, and they were once asset light, intangible based companies, and now they've become asset heavy, and they spend between 60% and 120% of their operating income on physical equipment. And that really is stunning.

Even so, there seems to be this anxiousness about AI. We keep getting questions like, "When are we going to see it?" And when I tried to explain sometimes, or we feel that sometimes the answers are unsatisfying because maybe some new user is using AI, but it's become some sort of effortless adoption. For example, when was your last cash transaction in a store, or when did you start using Visa cards to pay for \$2 items? Everyone will have their own answer, but in my guess it is for most people in America sort of 5-10 years ago. However, the data suggests that AI chats, AI customer service agents, coding, translation, they're all being used now at a massively accelerating rate. So a lot of times new technologies, you don't quite feel them when they're happening. But what we wanted to highlight was those applications are happening now and that AI spending is happening now.

John Przygocki: So, Matt, in the colossal tech piece, one analogy that you highlight is this investment made by the United States when it introduced the Marshall Plan. Why do you think that's a useful comparison?

Matt Moberg: Well, the Marshall Plan is a very famous plan that helped rebuild Europe. In today's dollars, it was about \$32 to \$37 billion a year. And it's been studied vigorously by economists. And estimates are that every dollar given to Western Europe created about 3-4 times of additional economic activity. The spending over the last 12 months for AI has been about \$280 billion. It's about eight times what the Marshall Plan was. Now we don't know what the multiplier will be, but we feel confident that there will be a multiplier.

And so it follows that this spending is going to massively change our economy. Certainly it will produce a lot of energy production, and it will have a lot of other auxiliary economic activity.

John Przygocki: Another highlight from that piece is the investment made by Apple in China. What's so interesting there?

Matt Moberg: We had just finished reading the book *Apple in China* by Patrick McGee, and he highlighted that, in 2016, Apple spent about \$55 billion per year to manufacture in China.

Now we don't have any data here, but in my opinion, there's a high correlation between that spending and China becoming a leader in world production of things. We would argue that those two things were linked. Even though data of any sort out of China is limited, we suspect China had additional economic gains from that investment. So we think the anecdotal evidence is quite clear on that point.

John Przygocki: All right, Matt, I have one final question for you on the colossal tech piece. You referenced the expanded role of the corporate executive in today's world. Do you have any interesting thought for our listeners on that expanded role?

Matt Moberg: Yes, this is a bit of a side point, and it's beyond my investing scope. But I did find it interesting and my team found it interesting that unlike the Marshall Plan, which was instituted by the elected officials of the United States, and it was designed not only for the generosity of repairing broken nations after a devastating war, World War II, but it was also designed around politics. The United States wanted a robust constituency of democratic trading partners.

The Apple investment, by contrast, was made by a single company and really a single CEO and board. There was little to my knowledge of broader political gain strategically for the United States, and these decisions were not made by elected officials. And it highlights just how big these colossal companies have become. They really are the size of nations. And this trend is only gaining momentum. Arguably, the CEOs of all the trillion-dollar companies now wield as much geopolitical power as elected officials. And you can see this in their overtures to the current administration. So we just found that very interesting.

John Przygocki: All right, Matt, I'm just taking a look at my notes here, and I have a note on something you said earlier. You mentioned clients used to find your answer about when we'll see the results of AI somewhat unsatisfying. How did you answer that question previously?

Matt Moberg: Well, I should mention why I felt that it was unsatisfying, and perhaps it was not for everybody. But from my point of view, there is general agreement that after we make all the spend, the applications of AI will come, and there's concern around when is this going to come. And certainly today it's being used to write code. It's used to satisfy customer service questions and answer many questions that we have. It can translate languages in real time between people. It can drive taxis around busy streets without incident.

And this continues to happen at an accelerating rate. For example, in 2024, it took ChatGPT 12 months to triple its weekly active users from 100 to 300 million. And in 2025, it took about three months to nearly triple users, to grow about 2.5x, when they went from 300 to 700 million in the first three months of the year. And ChatGPT queries have doubled from four to eight per day. So a lot of times, and it comes back to that point, a lot of times when we talk about massive change happening, I think it's very typical for people if a lot of tech does not have that a-ha moment, but instead we adopt it so effortlessly that we don't always notice how much we're using it.

I will say one last piece is cloud revenues, which is, you know, AWS, Azure, ChatGPT, etc., is almost at a \$300 billion run rate. And that is also accelerating, in general being supply and not demand constrained.

So there certainly seems to be very good demand in these applications, even though we're still quite early in the technology and even though from our point of view, oftentimes many people don't even notice that they're using this technology more than they were before.

John Przygocki: So, Matt, as we come to the close of today's conversation, I'm wondering if there are any final thoughts that you'd like to leave our listeners with today?

Matt Moberg: Well, in this conversation, we spoke a lot about history and what a dynamic time we are in economically. And it's been super interesting. And I think that matters right now because we are in a period of time where we are creating the next generation companies really for the next hundred years. And that means there are fantastic opportunities for historically large asymmetrical returns in the marketplace.

I would argue that what you, as a potential investor, would want to do would be to find a manager that understands this period of time that we're going through, understands this level of change, has a process that can take advantage of this change, and, of course, has a prospectus that allows that team to take full advantage in all the industries and all the sectors to take advantage of the change that's occurring today.

So that's really what we wanted to leave people with. Other than that, I just really appreciate your time.

John Przygocki: That's terrific. And Matt, thank you for sharing your inspiring insight with us today. To all of our listeners, thank you for spending your valuable time with us for today's exciting and informative update. If you would like to hear more Talking Markets with Franklin Templeton, please visit our archive of previous episodes and subscribe on Apple Podcasts, Google Podcasts, Spotify or just about any other major podcast provider.

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Endnotes

The "Magnificent Seven" are Alphabet, Amazon, Apple, Meta, Microsoft, Nvidia and Tesla.

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